

SOFTWARE REQUIREMENTS SPECIFICATION

Internal eBay Helpdesk & Customer Support
Management Platform

System Name: eBay Omni-Desk Platform

Version: 1.0

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Prepared by: Solution Architecture Team

Confidentiality: Internal Use Only

2. Revision History

Version	Date	Author	Description of Changes
0.1	2026-04-10	Solution Architecture Team	Initial draft creation and high-level scope definition.
0.5	2026-05-02	Solution Architecture Team	Added detailed functional requirements, user stories, and initial DB schema.
0.8	2026-05-20	Engineering & QA Leads	Incorporated API specifications, workflow diagrams, and acceptance criteria.
1.0	2026-06-12	Solution Architecture Team	Final baseline approved for Engineering; updated Multi-Marketplace phase notes.

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3. Executive Summary

This Software Requirements Specification (SRS) defines the comprehensive technical and functional blueprint for the eBay Omni-Desk Platform—an Internal eBay Helpdesk & Customer Support Management Platform. The platform serves to modernize and unify customer support operations across a heavily fragmented portfolio of eBay seller accounts.

In the current operating model, support agents manage buyer communications by manually logging into individual eBay storefronts. This disjointed workflow results in severe inefficiencies, prolonged First Response Times (FRT), dropped messages, and an absolute lack of managerial oversight. The eBay Omni-Desk Platform resolves these critical pain points by ingesting all buyer messages via the eBay API into a centralized, PostgreSQL-backed data store and presenting them through a high-performance React frontend.

The platform introduces a WhatsApp-style conversation interface, fundamentally shifting the paradigm of buyer-seller communication from disjointed "emails" to persistent, threaded timelines. By empowering agents with rapid-reply capabilities, rich text templates, and explicit ticket assignment, the platform promises to dramatically decrease operational overhead.

The architectural foundation utilizes FastAPI (Python) for robust REST API delivery, Celery and Redis for asynchronous message synchronization, and Docker Compose for predictable, containerized deployments. This document scopes Phase 1 (Minimum Viable Product) while establishing the extensibility hooks required for Phase 2 (AI/SLA) and Phase 3 (Multi-Marketplace).

Objective Category	Key Outcome Expected
Operational Efficiency	Eliminate 100% of manual credential sharing and account switching.
Response Metrics	Reduce average First Response Time (FRT) by an estimated 45%.
Accountability	Establish 100% traceability for ticket ownership, reply actions, and status changes.
Scalability	Support seamless onboarding of new eBay accounts without adding proportional support headcount.

4. Business Overview

Current-State Problems

The organization currently operates over a dozen high-volume eBay seller accounts. Support staff utilize browser profiles and shared password vaults to authenticate into each account directly on eBay.com. This results in significant context-switching, accidental concurrent replies by multiple agents to the same buyer, unmeasured SLA breaches, and a lack of standardized response templates.

Target-State Vision

The target state is a singular web application where all incoming messages from all connected eBay accounts are pulled into a unified queue. Support agents log in via corporate credentials, are assigned conversations, and reply directly from the platform. Operational managers possess real-time dashboard visibility into queue depths, agent productivity, and bottleneck areas.

Business Drivers

Primary drivers include maintaining Top Rated Seller status on eBay (which requires rapid response times and defect minimization), reducing labor costs per resolved ticket, and standardizing the brand voice across all marketplace channels.

Operating Assumptions

- eBay API limits are sufficient to support 5-minute polling intervals across all connected accounts.
- Support teams will operate primarily on desktop/laptop environments (responsive mobile design is secondary).
- The business holds proper administrative rights to all eBay accounts required for OAuth onboarding.

Dependencies & Risks

- **Dependency:** eBay REST API stability and feature parity with their native messaging UI.
- **Risk:** Changes to eBay's OAuth token expiration policies could disrupt background synchronization. Mitigation: Automated alert notifications for token failures.
- **Risk:** High message volume during peak seasons (Q4) may strain the Celery workers. Mitigation: Horizontally scalable worker topology.

5. Project Goals

The successful delivery of this platform will be measured against the following explicit goals:

1. **Centralize Support Operations:** Unify messages from all eBay seller accounts into a single interface by MVP launch.
2. **Reduce Response Times:** Decrease average response time from current baselines to under 2 hours during business hours.
3. **Eliminate Account Switching:** Agents will achieve a 0% reliance on direct eBay login for messaging purposes.
4. **Improve Team Visibility:** Provide managers with real-time analytics on queue volumes and individual agent output.
5. **Track Ownership:** Enforce strict assignment rules so exactly one agent is accountable for a conversation at any given time.
6. **Provide Management Reporting:** Deliver automated reporting on categorization trends to identify upstream supply chain issues.

7. Maintain Communication History: Persist a comprehensive, immutable audit log of all interactions and system events.

6. Scope

This SRS bounds the technical and functional delivery of the Internal eBay Helpdesk Platform. The document exhaustively details the Phase 1 (MVP) implementation which is required to achieve initial operational capability. It also outlines the architectural provisions necessary to ensure Phase 2 (Advanced Analytics, AI, SLAs) and Phase 3 (Multi-Marketplace) can be seamlessly integrated without requiring core database or API rewrites.

7. In Scope

The Phase 1 (MVP) delivery is strictly scoped to include the following modules:

- **Authentication:** Secure login, logout, JWT session management, and self-service password reset.
- **RBAC:** Hierarchical roles (Admin, Operations Manager, Support Agent).
- **User Management:** Administrative tools to create, update, disable users, and force password resets.
- **eBay Account Management:** OAuth 2.0 integration to connect/disconnect accounts, automated token refresh, and connection health monitoring.
- **Message Synchronization:** Celery-based background polling every 5 minutes, new message detection, conversation threading, and database persistence.
- **Conversation Management:** Centralized list view, threaded WhatsApp-style detail view, global search, and filtering (by status, account, assignment).
- **Reply Management:** Text reply dispatching, draft saving, automatic retry logic for failed API calls, and delivery state tracking.
- **Template Management:** Creation, editing, deletion, and categorization of rich-text templates with one-click insertion.
- **Assignment:** Manual assignment and reassignment of conversations to active agents, backed by assignment history logs.
- **Categories:** Tagging conversations with predefined dispositions (Order Status, Return Request, Cancellation, Refund Inquiry, Product Inquiry, General Support).
- **Notifications:** Real-time in-app alerts for assignments, failed replies, and account connection drops.
- **Dashboard:** High-level metrics showing open conversations, assigned queues, resolution counts, and basic productivity metrics.
- **Audit Logging:** Granular tracking of logins, logouts, assignments, category shifts, replies sent, and template usage.

8. Out of Scope

The following functionalities are explicitly excluded from the Phase 1 MVP release to maintain timeline constraints:

- Automated conversation categorization using Artificial Intelligence or NLP.
- Automated, rules-based ticket assignment (Round-robin, skills-based routing).
- Strict SLA breach alerts and complex SLA timer logic (business hours pausing, holiday calendars).
- Internal, private conversational notes and agent @mentions.
- AI-generated reply suggestions based on historical data.
- Integration with any marketplaces other than eBay (e.g., Amazon, Etsy, Walmart).
- Custom drag-and-drop report builders.
- Handling of native eBay cases/disputes beyond standard messages.

9. Stakeholders

Stakeholder Role	Responsibilities & Interests
VP of Customer Experience	Executive Sponsor. Interested in overall ROI, customer satisfaction metrics, and strategic alignment.
Director of Support Operations	Product Owner. Drives functional requirements, prioritization, and business process alignment.
Operations Managers	Primary management users. Require deep visibility, dashboard metrics, and workload balancing tools.
Support Agents	Primary end users. Require a fast, reliable, intuitive UI that minimizes clicks and friction.
Solution Architecture Team	Defines technical blueprints, ensures scalability, security compliance, and robust integration patterns.
Development & QA Team	Responsible for the execution, automated testing, and deployment of the specified requirements.

10. User Roles

Role	Description	System Permissions
Admin	IT and System Administrator. Full platform control.	All capabilities. Exclusive access to User Management, global platform configurations, and deep Audit Logs.
Operations Manager	Team leader managing day-to-day queue health.	Manage eBay connections, manage global templates, view all dashboards, force-reassign any conversation, view agent metrics.
Support Agent	Frontline staff focused on resolving buyer inquiries.	View assigned and unassigned conversations, send replies, apply categories, utilize templates, update conversation statuses. Cannot delete data.
System Actor (Sync Engine)	Background worker executing automated tasks.	Read/Write to conversations, messages, audit logs. Uses internal service credentials.

11. Functional Requirements

11.1 Authentication

The authentication module provides secure access control using JSON Web Tokens (JWT). It ensures that only authorized staff can access customer data. The module supports login, logout, and self-service password recovery.

User Stories:

- As a Support Agent, I want to log in securely with my email and password so that I can access the system.
- As a User, I want my session to auto-refresh while I am active so that I am not abruptly logged out while typing a reply.
- As a User who forgot their password, I want to request a reset link via email so I can regain access independently.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-AUTH-001	Auth	System shall authenticate users via Email and Password, issuing short-lived access JWTs and long-lived refresh tokens.	Critical	Phase 1	Access token: 15 mins. Refresh token: 8 hours.
FR-AUTH-002	Auth	System shall provide a password reset flow generating a time-limited, single-use token sent via email.	High	Phase 1	Token expires in 1 hour.
FR-AUTH-003	Auth	System shall invalidate the refresh token on explicit logout.	High	Phase 1	Redis blocklist utilized for token revocation.

11.2 Role Based Access Control

RBAC restricts network and application access based on the roles of individual users. The system hardcodes three specific roles for MVP, enforced at the API routing layer.

User Stories:

- As an Admin, I want to ensure Agents cannot access the user creation screens to maintain security.
- As an Operations Manager, I want to access dashboard analytics that Agents cannot see.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-RBAC-001	RBAC	System shall restrict access to User Management APIs exclusively to the Admin role.	Critical	Phase 1	Return 403 Forbidden for unauthorized roles.
FR-RBAC-002	RBAC	System shall allow Admin and Operations Manager roles to manage eBay Account connections.	High	Phase 1	UI must hide these navigation elements from Agents.

11.3 User Management

The User Management module provides administrative functions to provision, update, and de-provision employee access to the platform.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-USER-001	User Mgt	System shall allow Admins to create new user accounts, defining Email, Name, and Role.	High	Phase 1	Triggers welcome email with password setup link.
FR-USER-002	User Mgt	System shall allow Admins to update user details and roles.	High	Phase 1	Role changes take effect on next token refresh.
FR-USER-003	User Mgt	System shall allow Admins to soft-disable users, immediately revoking access.	Critical	Phase 1	Disabled users cannot log in; existing tokens blacklisted.

11.4 eBay Account Management

This module handles the OAuth 2.0 lifecycle with eBay's APIs, allowing the platform to act on behalf of multiple seller accounts.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-EBY-001	eBay Mgt	System shall implement the eBay OAuth authorization code grant flow to connect seller accounts.	Critical	Phase 1	Persist user-level refresh tokens securely.
FR-EBY-002	eBay Mgt	System shall automatically refresh short-lived eBay access tokens using the stored refresh token prior to API calls.	Critical	Phase 1	Handle token rotation silently.
FR-EBY-003	eBay Mgt	System shall monitor connection health and flag accounts requiring manual re-authentication.	High	Phase 1	Generate system notification on persistent refresh failure.

11.5 Message Synchronization

The sync engine is the background heartbeat of the application, responsible for polling eBay APIs and persisting state locally.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-SYNC-001	Sync	System shall poll the eBay Messages API for every connected, healthy account at a configurable interval (default 5 minutes).	Critical	Phase 1	Use Celery beat scheduler.
FR-SYNC-002	Sync	System shall parse incoming messages, grouping them into Conversations based on eBay Item ID and Buyer ID.	Critical	Phase 1	If conversation exists, append message. If not, create new.
FR-SYNC-003	Sync	System shall drop duplicate messages ingested across overlapping poll windows.	High	Phase 1	Unique constraint on eBay Message ID.

11.6 Conversation Management

This is the primary workspace for agents. It mimics a modern chat interface rather than a traditional email client.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-CONV-001	Conv	System shall present a master-detail view of conversations, sortable by most recent message timestamp.	Critical	Phase 1	Unread messages indicated visually.
FR-CONV-002	Conv	System shall allow filtering conversations by Status (Open, Resolved), eBay Account, and Assignee.	High	Phase 1	Filters are multi-select where applicable.
FR-CONV-003	Conv	System shall allow agents to mark an Open conversation as Resolved.	High	Phase 1	Resolved conversations drop from the default 'Open' view.

11.7 Reply Management

Handles outbound communications from the agent through the system to the buyer via eBay's API.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-REPLY-001	Reply	System shall allow agents to compose and send text replies directly within the conversation thread.	Critical	Phase 1	Max characters matching eBay limits (typically 2000).
FR-REPLY-002	Reply	System shall persist the outbound message locally as 'PENDING' before dispatching to eBay.	Critical	Phase 1	Prevents data loss on API timeout.
FR-REPLY-003	Reply	System shall track and display delivery status visually (Pending, Sent, Failed).	High	Phase 1	Use checkmark icons akin to WhatsApp.

11.8 Template Management

Enables standardization of responses to common queries.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-TPL-001	Template	System shall allow Managers/Admins to CRUD pre-written text templates.	High	Phase 1	Supports basic dynamic variables (e.g., {{buyer_name}}).
FR-TPL-002	Template	System shall group templates by Category for easy retrieval.	Medium	Phase 1	UI dropdown or modal selection.
FR-TPL-003	Template	System shall allow agents to insert templates into the active reply box with a single click.	High	Phase 1	Appends to existing typed text.

11.9 Assignment & Ownership

Ensures clear accountability and prevents duplicate agent effort.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-ASG-001	Assign	System shall allow agents to manually claim (assign to self) an unassigned conversation.	High	Phase 1	Updates UI for all active users via WebSocket/Polling.
FR-ASG-002	Assign	System shall allow Managers to explicitly assign a conversation to any specific active agent.	High	Phase 1	Generates notification for the targeted agent.

11.10 Categories

Provides basic taxonomy for inquiry types.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-CAT-001	Category	System shall support application of predefined categories to a conversation.	Medium	Phase 1	Order Status, Return Request, Cancellation, Refund, Product Inquiry, General Support.
FR-CAT-002	Category	System shall allow changing the category at any point in the conversation lifecycle.	Medium	Phase 1	Logged in Audit table.

11.11 Notifications

Keeps users informed of critical state changes.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-NOTIF-001	Notif	System shall generate an in-app notification when an agent is assigned a ticket by someone else.	High	Phase 1	Appears in a notification bell dropdown.
FR-NOTIF-002	Notif	System shall generate a notification if an agent's outbound reply permanently fails delivery.	High	Phase 1	High priority alert.

11.12 Dashboard & Operational Metrics

Provides heads-up display of platform health.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-DASH-001	Dash	System shall display global counts of Open, Unassigned, and Resolved conversations (Last 24h).	High	Phase 1	Updates on page load / hard refresh.
FR-DASH-002	Dash	System shall display personal metrics for logged-in agents (Assigned to Me, Replies Sent Today).	Medium	Phase 1	Personalized view context.

11.13 Audit Logging

Ensures complete compliance and operational forensics.

ID	Module	Requirement	Priority	Phase	Acceptance Notes
FR-AUD-001	Audit	System shall log all state changes to conversations (Status, Category, Assignment) with User ID and Timestamp.	Critical	Phase 1	Immutable database records.
FR-AUD-002	Audit	System shall log authentication events (Login success, Login failure, Logout).	High	Phase 1	Captures IP address and timestamp.

11.14 Phase 2 Functional Additions

Future features to automate triage and monitor service levels.

- **FR-AI-001 (AI Categorization):** System shall automatically suggest or apply categories based on message intent using NLP.
- **FR-SLA-001 (SLA Tracking):** System shall visually flag conversations approaching First Response SLA limits.
- **FR-BLK-001 (Bulk Actions):** System shall allow managers to select multiple conversations to bulk-resolve or bulk-assign.

11.15 Phase 3 Functional Additions

Future features for enterprise collaboration and multi-channel expansion.

- **FR-COL-001 (Internal Notes):** System shall allow agents to leave private, internal-only comments on a thread.
- **FR-COL-002 (@Mentions):** System shall support @tagging other users in internal notes to trigger notifications.
- **FR-MKT-001 (Multi-Marketplace):** System architecture shall abstract marketplace APIs to support Amazon, Walmart, and Etsy messaging.

12. Non-Functional Requirements

Category	ID	Requirement	Target / Metric	Verification Method
Performance	NFR-PERF-01	Initial page load (Dashboard/List) under normal network conditions.	< 3 seconds	Lighthouse / Chrome DevTools
Performance	NFR-PERF-02	Standard API response time (Read/Write operations, excl. 3rd party API calls).	< 500ms at 95th percentile	APM Tooling / JMeter
Availability	NFR-AVAIL-01	System uptime targeting standard business operating hours.	99.5%	Synthetics / Pingdom
Scalability	NFR-SCAL-01	Support concurrency limits without degradation.	25 concurrent users; 400 new msgs/day	Load Testing
Security	NFR-SEC-01	Passwords and sensitive tokens must be protected at rest.	bcrypt (pw); AES-256 (tokens)	Code Review / DB Inspection
Reliability	NFR-REL-01	Background jobs interacting with external APIs must gracefully handle network failures.	Max 3 retries with exponential backoff; dead-letter queue	Chaos Engineering / Log Analysis
Browser Support	NFR-CMP-01	Frontend compatibility with modern browsers.	Chrome, Edge, Firefox (Last 2 versions)	Manual QA
Maintainability	NFR-MNT-01	System database schema must be version-controlled.	Alembic Migrations implemented	CI/CD Pipeline Checks

13. Business Rules

- **BR-001 (Strict Ownership):** A conversation can only be assigned to a single Support Agent at any given time.
- **BR-002 (Reassignment Lock):** If a conversation is assigned to Agent A, Agent B cannot submit a reply unless they first explicitly reassign the conversation to themselves (logging the takeover).
- **BR-003 (Reply Idempotency):** All outbound API requests to eBay for message submission must include a unique idempotency key to prevent accidental duplicate messages during network retries.
- **BR-004 (SLA Start):** SLA timers implicitly begin at the exact database insertion timestamp of an inbound buyer message.
- **BR-005 (SLA Stop):** SLA timers implicitly stop when the system logs a 'SENT' status for an agent's outbound reply.

- **BR-006 (Duplicate Prevention):** The sync engine must drop incoming messages whose eBay native Message ID already exists in the database.
- **BR-007 (Audit Immutability):** Rows in the `AuditLog` table are append-only. UPDATE and DELETE operations are strictly forbidden at the database level.
- **BR-008 (User Disabling):** Disabling a user immediately blacklists their active JWTs, effectively terminating session access on the next API call.
- **BR-009 (Soft Deletion):** Users, Templates, and connected eBay Accounts are never hard-deleted; they are marked via a boolean `is_active` or `is_deleted` flag.
- **BR-010 (Token Refresh):** The system must attempt to refresh an eBay OAuth token 5 minutes before its scheduled expiration.
- **BR-011 (Notification Deduplication):** Identical system notifications (e.g., repeating sync failures for the same account) within a 1-hour window should be aggregated rather than spammed.
- **BR-012 (Category Inheritance):** New messages appended to an existing conversation inherit the conversation's currently assigned category.
- **BR-013 (Template Permissions):** Support Agents have Read-Only access to global templates. Only Managers and Admins possess Write capabilities.
- **BR-014 (Auto-Resolution):** A conversation must NOT be automatically resolved upon reply; the agent must explicitly change the status.
- **BR-015 (Account Disconnect):** Disconnecting an eBay account ceases future message syncs but preserves historical conversation data indefinitely.

14. System Architecture

The Omni-Desk platform utilizes a modern, modular monolithic architecture designed for eventual microservice extraction. The backend API is constructed with FastAPI (Python) to leverage high-performance asynchronous IO. The persistence layer relies on PostgreSQL for relational integrity, while Redis acts as a high-speed broker for background task queues and token blacklisting.

The Sync Engine operates as a cluster of Celery workers. These workers run periodically to fetch data from eBay, transforming external proprietary formats into an abstracted internal representation. This extensibility layer ensures that integrating Amazon or Walmart in Phase 3 requires only a new worker parser, while the core database schema and frontend remain largely untouched.

Component Responsibility Table

Component	Technology	Responsibility
Frontend SPA	React, TypeScript	Client rendering, state management, routing, WebSocket consumption.
Backend API	FastAPI, Python	RESTful endpoints, RBAC enforcement, DB CRUD, business logic validation.
Database	PostgreSQL	ACID compliance, persistent data storage, relational integrity.
Message Broker	Redis	Celery task queuing, JWT blacklisting, rate limit counters.
Background Workers	Celery	eBay API polling, heavy data parsing, retry handling, email dispatch.

14.1 System Architecture Diagram

```
graph TD
    subgraph External
        eBay[eBay API]
    end
    subgraph Omni-Desk_Backend_Environment [Omni-Desk Backend Environment]
        Nginx[Nginx Reverse Proxy]
        API[FastAPI Backend Server]
        CelerySync[Celery Sync Service]
        CeleryTask[Celery Task Worker]
        DB[(PostgreSQL)]
        Cache[(Redis Cache/Broker)]
    end
    subgraph Client_Environment [Client Environment]
        React[React SPA UI]
    end
    React -->|REST / HTTPS| Nginx
    Nginx --> API
    API <-->|SQLAlchemy| DB
    API <-->|Auth / State| Cache
    CelerySync <-->|OAuth / JSON| eBay
    CelerySync -->|Ingest Data| DB
    CelerySync <-->|Job Queue| Cache
    CeleryTask -->|Outbound Replies| eBay
    eBay API -->|Enqueue Task| Cache
```

14.2 Message Flow Diagram (Inbound)

```
sequenceDiagram
    participant Buyer
    participant eBay
    participant SyncWorker
    participant DB
    participant API
    participant UI
    Buyer->>eBay: Sends message via eBay Portal loop Every 5 Mins
    SyncWorker->>eBay: GET /messages
    eBay-->>SyncWorker: JSON Message List
    SyncWorker->>SyncWorker: Deduplicate & Parse
    SyncWorker->>DB: UPSERT Conversation & Message
    DB-->>SyncWorker: Success
    SyncWorker->>API: Webhook/Event Trigger
    API->>UI: Websocket Notification (New Message)
    UI->>API: GET /conversations/update
    API-->>UI: Updated Payload
```

14.3 Reply Flow Diagram (Outbound)

```
sequenceDiagram
    participant Agent
    participant UI
    participant API
    participant DB
    participant TaskWorker
    participant eBay

    Agent->>UI: Types reply and clicks Send
    UI->>API: POST /conversations/{id}/reply
    API->>DB: INSERT Message (Status: PENDING)
    API->>TaskWorker: Enqueue Reply Task (Async)
    API-->>UI: 202 Accepted (Pending UI state)
    TaskWorker->>eBay: POST /send_message
    alt API Success
        eBay-->>TaskWorker: 200 OK
    else API Failure
        eBay-->>TaskWorker: 500 Error
    end
    TaskWorker->>DB: UPDATE Message (Status: SENT)
    else API Failure
        eBay-->>TaskWorker: 500 Error
    end
    TaskWorker->>TaskWorker: Retry Logic (Max 3)
    TaskWorker->>DB: UPDATE Message (Status: FAILED)
    end
```

14.4 Deployment Diagram

```
graph LR
    subgraph Docker
        subgraph Host
            Proxy[Proxy]
        end
        subgraph UI[React Container]
            UI
        end
        subgraph API[FastAPI Container]
            API
        end
        subgraph CeleryB[Celery Beat]
            CeleryB
        end
        subgraph CeleryW[Celery Worker]
            CeleryW
        end
        subgraph DB[(PostgreSQL 15)]
            DB
        end
        subgraph Redis[(Redis 7)]
            Redis
        end
    end
    Internet((Internet)) --> Proxy
    Proxy --> UI
    Proxy --> API
    API --> DB
    API --> Redis
    CeleryB --> Redis
    CeleryW --> Redis
    CeleryW --> DB
```

14.5 RBAC Diagram

```
flowchart TD
    User([System User]) --> Admin[Admin]
    User --> Mgr[Operations Manager]
    User --> Agt[Support Agent]
    Admin --> U[Users]
    Admin --> V[Views]
    Admin --> A[Audit Logs]
    Admin --> I1[Inherits]
    Mgr --> E[eBay Accounts]
    Mgr --> T[Templates]
    Mgr --> O[Overrides]
    C[Conversations] --> I2[Inherits]
    Agt --> R1[Replies to]
    C --> R2[Reads]
    T --> R3[Reads]
    D[Personal Dashboard]
```

15. Database Design

The database relies on PostgreSQL utilizing SQLAlchemy ORM. Soft deletes are preferred over hard deletes. All primary keys utilize UUID v4 to prevent enumeration attacks and support complex distributed architectures if needed later.

15.1 Entity Relationship Diagram

```
erDiagram
    USER {
        uuid id PK
        string email
        string password_hash
        uuid role_id FK
        boolean is_active
    }
    ROLE {
        uuid id PK
        string name
    }
    EBAY_ACCOUNT {
        uuid id PK
        string seller_username
        string access_token
        string refresh_token
        boolean is_active
    }
    CONVERSATION {
        uuid id PK
        uuid ebay_account_id FK
        string external_thread_id
        string buyer_username
        string status
        uuid assigned_user_id FK
    }
    MESSAGE {
        uuid id PK
        uuid conversation_id FK
        string external_message_id
        string direction
        string content
        string status
        timestamp timestamp
        created_at
    }
    TEMPLATE {
        uuid id PK
        string name
        string content
        uuid category_id FK
    }
    CATEGORY {
        uuid id PK
        string name
    }
    AUDIT_LOG {
        uuid id PK
        uuid user_id FK
        string action
        string entity_type
        uuid entity_id
        timestamp timestamp
        created_at
    }
    USER ||--o{ ROLE : contains
    EBAY_ACCOUNT ||--o{ CONVERSATION : receives
    CONVERSATION ||--o{ MESSAGE : contains
    USER ||--o{ CONVERSATION : manages
    CATEGORY ||--o{ CONVERSATION : categorizes
    CATEGORY ||--o{ TEMPLATE : organizes
    USER ||--o{ AUDIT_LOG : generates
```

15.2 Entity Descriptions & Data Dictionaries

Table: User

Field Name	Data Type	Required	Description
id	UUID	Yes	Primary Key.
email	VARCHAR(255)	Yes	Unique identifier, used for login.
password_hash	VARCHAR(255)	Yes	Bcrypt hashed password.
role_id	UUID	Yes	Foreign Key to Role table.
first_name	VARCHAR(100)	Yes	Agent's first name.
last_name	VARCHAR(100)	Yes	Agent's last name.
is_active	BOOLEAN	Yes	Default TRUE. Soft delete flag.
created_at	TIMESTAMPTZ	Yes	Record creation timestamp.

Table: Role

Field Name	Data Type	Required	Description
id	UUID	Yes	Primary Key.
name	VARCHAR(50)	Yes	'Admin', 'Operations Manager', 'Support Agent'.

Table: EbayAccount

Field Name	Data Type	Required	Description
id	UUID	Yes	Primary Key.
seller_username	VARCHAR(100)	Yes	eBay Handle (e.g., super_electronics_store).
access_token	TEXT	Yes	Encrypted short-lived token.
refresh_token	TEXT	Yes	Encrypted long-lived token.
token_expires_at	TIMESTAMPTZ	Yes	Timestamp of access token expiration.
is_active	BOOLEAN	Yes	If false, sync engine skips this account.

Table: Conversation

Field Name	Data Type	Required	Description
id	UUID	Yes	Primary Key.
ebay_account_id	UUID	Yes	Foreign Key indicating origin account.
external_thread_id	VARCHAR(255)	No	eBay's native group ID (if applicable).
buyer_username	VARCHAR(100)	Yes	Buyer's eBay handle.
item_id	VARCHAR(100)	No	Associated eBay Listing ID.
status	VARCHAR(50)	Yes	'OPEN', 'PENDING', 'RESOLVED'.
assigned_user_id	UUID	No	Nullable FK to User handling the ticket.
category_id	UUID	No	Nullable FK to Category.
updated_at	TIMESTAMPTZ	Yes	Last message/action timestamp (for sorting).

Table: Message

Field Name	Data Type	Required	Description
id	UUID	Yes	Primary Key.
conversation_id	UUID	Yes	Foreign Key to Conversation.
external_message_id	VARCHAR(255)	Yes	eBay Native Message ID (Unique).
direction	VARCHAR(10)	Yes	'INBOUND' (Buyer) or 'OUTBOUND' (Agent).
content	TEXT	Yes	Raw message text.
status	VARCHAR(50)	Yes	'RECEIVED', 'PENDING_SEND', 'SENT', 'FAILED'.
created_at	TIMESTAMPTZ	Yes	When message was originated.

Table: AuditLog

Field Name	Data Type	Required	Description
id	UUID	Yes	Primary Key.
user_id	UUID	Yes	Actor triggering the event.
action	VARCHAR(100)	Yes	e.g., 'CONVERSATION_ASSIGNED', 'REPLY_SENT'.
entity_type	VARCHAR(50)	Yes	'USER', 'CONVERSATION', 'ACCOUNT'.
entity_id	UUID	Yes	UUID of the affected record.
payload	JSONB	No	Optional snapshot of data changes.
created_at	TIMESTAMPTZ	Yes	Immutable timestamp.

16. API Specifications

The REST API strictly utilizes JSON payloads. Authentication is managed via HTTP Authorization Header (Bearer token). All API endpoints return standardized error formats and appropriate HTTP status codes (200, 201, 202, 400, 401, 403, 404, 500).

16.1 Authentication APIs

Method	Route	Purpose	Roles
POST	/api/v1/auth/login	Exchanges credentials for JWTs.	Public
POST	/api/v1/auth/logout	Invalidates active refresh token.	All Authenticated
POST	/api/v1/auth/refresh	Provides new access token.	Valid Refresh Token
POST	/api/v1/auth/forgot-password	Triggers reset email.	Public

16.2 User APIs

Method	Route	Purpose	Roles
GET	/api/v1/users	List platform users.	Admin, Manager
POST	/api/v1/users	Create a new user.	Admin
PATCH	/api/v1/users/{id}	Update user details/status.	Admin

16.3 eBay Account APIs

Method	Route	Purpose	Roles
GET	/api/v1/ebay-accounts	List connected accounts & health.	Admin, Manager
POST	/api/v1/ebay-accounts/connect	Initiate OAuth flow.	Admin, Manager
POST	/api/v1/ebay-accounts/{id}/refresh	Force manual token refresh.	Admin, Manager

16.4 Conversation & Reply APIs

Method	Route	Purpose	Roles
GET	/api/v1/conversations	List/search/filter conversations.	All
GET	/api/v1/conversations/{id}	Get full message thread.	All
PATCH	/api/v1/conversations/{id}/assign	Assign thread to user.	All
PATCH	/api/v1/conversations/{id}/status	Mark Open/Resolved.	All
POST	/api/v1/conversations/{id}/replies	Enqueue outbound text reply.	All

Sample JSON: POST /api/v1/conversations/{id}/replies

```
// Request
{
  "content": "Hello, thank you for reaching out. Your order has been processed and
will ship tomorrow.",
  "template_used_id": "c1234567-89ab-cdef-0123-456789abcdef",
  "idempotency_key": "req-98765-abcde-12345"
}

// Response (202 Accepted)
{
  "status": "success",
  "data": {
    "message_id": "d9876543-210f-edcb-a987-6543210fedcb",
    "delivery_status": "PENDING_SEND",
    "created_at": "2026-06-12T14:32:00Z"
  }
}
```

16.5 Template APIs

Method	Route	Purpose	Roles
GET	/api/v1/templates	List active templates.	All
POST	/api/v1/templates	Create new template.	Admin, Manager

16.6 Analytics APIs

Method	Route	Purpose	Roles
GET	/api/v1/analytics/overview	Global ticket metrics.	Admin, Manager
GET	/api/v1/analytics/agents	Agent productivity stats.	Admin, Manager

17. UI/UX Specifications

The platform UI is optimized for high-frequency operational work rather than low-volume case management. The visual model emphasizes queue triage, fast reading, minimal clicks to reply, and continuous visibility into assignment, ownership, and conversation state. The information architecture follows a role-sensitive left navigation with context-aware content panes. The default experience uses a three-zone layout: navigation, list, and detail.

Accessibility expectations include WCAG 2.1 AA-inspired keyboard navigation, color contrast support, visible focus states, semantic labels on form fields, and screen-reader-friendly headings for major work areas. Error messaging must be inline and actionable. Search must support buyer name, item/order reference, and conversation identifier. Filters must remain sticky during the active session and support reset to default.

UX Area	Specification
Primary Navigation	Dashboard, Conversations, Templates, eBay Accounts, Users, Audit Logs, Analytics, Profile/Logout
Responsive Baseline	Desktop-first at 1366x768 minimum; tablet support optional for read-only manager dashboards in Phase 2
Search Behavior	Debounced search (300–500ms), server-side filtering for list results, client display of active filter chips
Status Semantics	Open = blue, Pending = amber, Resolved = green, Failed = red, Warning = orange, Disabled = gray
Conversation Threading	WhatsApp-style alternating message bubbles, timestamps, sender identity, delivery state, category, assignee
Empty States	Provide actionable CTAs such as “Connect eBay Account”, “Create Template”, or “No Results — Clear Filters”

17.1 Wireframe: Login Screen

```

+-----+
-+
|
|                                     eBay  Omni-Desk
|
|                                     Internal  Helpdesk  &  Support  Platform
|
+-----+
-+
|
|
|                                     [  Company  Logo  /  Product  Mark  ]
|
|
|
|
|                                     Email  Address      [_____]
|
|                                     Password             [_____]
|
|
|
|                                     [  Sign  In  ]           [  Forgot  Password  ]
|
|
|
|                                     Status:  Secure  login  via  JWT
|
+-----+
-+

```

```

+-----+
---+
| [Logo] Omni-Desk          | Search Buyer or Item #...          | [🔔 2] [JD] Log
Out |
|
+-----+
| NAVIGATION                | DASHBOARD    OVERVIEW
|
| -----
|
| 🏠 Dashboard              | [ Open ] [ Unassigned ] [ Resolved ] [ Avg FRT ]
|
| 💬 Conversations          | [ 142 ] [ 12 ]          [ 89 ]          [ 00:18 ]
|
| 📄 Templates
|
| ⚙ Accounts              | RECENT ACTIVITY
|
PERFORMANCE |
| 👤 Users                | -----
-- |
| 📈 Analytics            | Ticket #1245 -> Jane S.          John D: 45
replies|
| 📋 Audit                | Ticket #1244 -> John D.          Jane S: 42
replies|
|                          | Account Sync Warning: Store B    Mark T: 12
replies|
+
---+

```

17.3 Wireframe: Conversation List

[illegible]

```

+
----+
|  TICKET: #99321  |  Buyer:  User123  |  eBay Account:  Store A  |  Status  [  OPEN ▼  ]
|
+
----+
|  ASSIGNEE  [  Me ▼  ]  |  CATEGORY  [  Order Status ▼  ]  |  PRIORITY  [  Normal  ]
|
+-----+
----+
|          Buyer          10:45          AM:          Where          is          my          tracking          number?
|
|-----+
---|
|  Agent 10:49 AM: Hello User123, your order is in transit and should arrive tomorrow.
|
+-----+
---|
|  Internal      Status:      Last      Sync      10:50      AM      |  Delivery      State:      Sent
|
|
|
---|
|  Template ▼  ]  [  Save  Draft  ]  [  Attach  future  feature  disabled  ]
|
|
|  Reply
|
|  ] |
|
|  ] |
|
|  ] |
|
|  [ Send Reply ] [ Mark Resolved ]
|
+
----+

```

[illegible]

```
+-----+  
---+  
| USER MANAGEMENT [ + Create  
User ] |  
+-----+  
---+  
| Filter Role [All ▼] Status [Active ▼] Search [_____] |  
| _____ |  
+-----+  
---|  
| John Doe | john@company.com | Support Agent | Active | [Edit]  
| _____ |  
| Jane Lead | jane@company.com | Operations Manager | Active | [Edit]  
| _____ |  
| Admin One | admin@company.com | Admin | Active | [Edit]  
| _____ |  
  
-----  
---|  
| Selected User |  
| _____ |  
| Name [_____] Email [_____] Role [Support Agent ▼] |  
| _____ |  
| Active [x] Force Reset Password [ ] |  
| _____ |  
| [ Save Changes ] [ Disable User ] |  
| _____ |  
+  
---+
```



```
+-----+
|                                     ANALYTICS                                     | DASHBOARD
|
+-----+
| Date Range [Last 7 Days ▼]   Account [All ▼]   Agent [All ▼]   Export [CSV] [PDF]
|
|-----+
| KPI CARDS: [Volume 400] [Avg FRT 00:21] [Resolution Rate 82%] [Backlog 57]
|
|-----+
| Trend Chart Placeholder | Category Split Placeholder
|
|-----+
| Team Performance Table | SLA / Future Phase Widget
|
| John D.   45 replies   Avg 00:17 | First Response SLA Met: 93%
|
| Jane S.   42 replies   Avg 00:15 | Resolution SLA Met: 88%
|
+-----+
+-----+
```

18. Workflow Diagrams

The following BPMN-style workflows describe the operational sequences for message ingestion, triage, reply dispatch, assignment control, and onboarding. Each workflow identifies trigger conditions, core path, exception handling, and audit expectations.

18.1 New Message Sync

```
flowchart TD
  A[Scheduler Trigger] --> B[Load Active eBay Accounts]
  B --> C[Request Messages from eBay API]
  C --> D{API Success?}
  D -- No --> E[Record Sync Failure]
  E --> F[Increment Retry Counter]
  F --> G[Notify Manager if Threshold Reached]
  D -- Yes --> H[Normalize Payload]
  H --> I[Check Duplicate External Message ID]
  I --> J{Duplicate?}
  J -- Yes --> K[Skip Insert]
  J -- No --> L[Persist Message]
  L --> M[Update/Create Conversation]
  M --> N[Create Notification if Needed]
  N --> O[Write Audit Log]
```

Trigger: Celery Beat executes a scheduled sync per healthy connected eBay account. The worker authenticates using the stored seller token, fetches messages since the previous checkpoint, normalizes the payload, and performs duplicate detection using external message identifiers.

Exception path: If eBay is unavailable or a token is invalid, the job records a failed sync attempt, increments the retry counter, and generates an account health warning for managers after the configured threshold. Audit coverage includes job start, job completion, failure reason, and number of messages ingested.

18.2 Conversation Creation

```
flowchart TD
  A[Inbound Message Persisted] --> B[Find Conversation by Thread Key]
  B --> C{Found?}
  C -- Yes --> D[Append Message to Existing Conversation]
  C -- No --> E[Create Conversation]
  E --> F[Set Initial Status Open]
  F --> G[Apply Default Category General Support]
  D --> H[Update Last Message Timestamp]
  G --> H
  H --> I[Queue Unassigned Notification]
  I --> J[Write Audit Event]
```

Trigger: a newly ingested buyer message has been accepted into the database. The system resolves the thread using a deterministic conversation key derived from eBay account, buyer identifier, and marketplace conversation/thread ID.

If the conversation does not exist, the platform creates an Open conversation, defaults category to General Support, and records it as unassigned. If the conversation already exists, only metadata and last-message fields are updated. Audit events capture whether a conversation was created or appended.

18.3 Reply Sending

```
flowchart TD
  A[Agent Clicks Send] --> B[Validate RBAC and Payload]
  B --> C[Persist Reply as PENDING]
  C --> D[Publish Dispatch Job]
  D --> E[Call eBay Reply API]
  E --> F{Successful?}
  F -- Yes --> G[Update Message Status SENT]
  F -- No --> J[Retry up to 3 Times]
  G --> H[Update Conversation Last Reply Timestamp]
  J --> K{Still Failed?}
  K -- No --> G
  K -- Yes --> L[Update Status FAILED]
  H --> I[Write Audit Event]
  L --> M[Notify Assignee]
  M --> I
```

Trigger: an assigned agent submits an outbound reply. The request is validated for authorization, payload length, conversation eligibility, and idempotency key uniqueness before a local message record is created with status `PENDING`.

The local-first persistence pattern prevents data loss and allows robust retry processing. Permanent failure generates a high-priority notification and leaves a durable audit record of each send attempt and final disposition.

18.4 Assignment

```
flowchart TD
    A[Manager or Agent Claims Conversation] --> B[Check Current Assignee]
    B --> C{Unassigned?}
    C -- Yes --> D[Assign to Requesting User]
    C -- No --> E[Evaluate Override Permission]
    E --> F{Authorized?}
    F -- No --> G[Reject with 403]
    F -- Yes --> H[Reassign Conversation]
    D --> I[Create Assignment History]
    H --> I
    I --> J[Notify New Assignee]
    J --> K[Write Audit Event]
```

Trigger: either an agent self-claims an unassigned ticket or a manager assigns it to another user. Assignment rules enforce single active ownership and prevent silent overwrites of another agent's work unless performed by an authorized manager or admin.

18.5 Reassignment

```
flowchart TD
    A[Manager Initiates Reassignment] --> B[Capture Reason]
    B --> C[Validate Target User Active]
    C --> D[Update Conversation Assignee]
    D --> E[Close Previous Active Assignment History Row]
    E --> F[Create New Assignment History Row]
    F --> G[Notify New Assignee]
    G --> H[Write Audit Event with Old/New Values]
```

Reassignment is a controlled managerial action. The system must persist both the previous and new assignee, the user initiating the change, the reason for the change, and the timestamp. This supports accountability and later workload reporting.

18.6 Template Usage

```
flowchart TD
    A[Agent Opens Template Picker] --> B[Filter by Category or Search]
    B --> C[Select Template]
    C --> D[Render Variables]
    D --> E[Insert into Draft Area]
    E --> F[Agent Edits Optional Text]
    F --> G[Send or Save Draft]
    G --> H[Log Template Usage]
```

Template usage is an acceleration pattern, not a hard automation. Agents may search by category, choose a template, and optionally edit the text before sending. Usage must be logged regardless of whether the final text is edited, so managers can track value realization from standardized responses.

18.7 Notification Flow

```
flowchart TD
    A[Domain Event Occurs] --> B[Evaluate Notification Rules]
    B --> C{Eligible Recipient Exists?}
    C -- No --> D[End]
    C -- Yes --> E[Deduplicate Event]
    E --> F[Persist Notification as UNREAD]
    F --> G[Push UI Refresh / Poll Result]
    G --> H[User Opens Notification Center]
    H --> I[Mark as READ]
    I --> J[Write Audit Event]
```

Notifications are generated from durable domain events such as assignment changes, failed replies, or account health degradation. The engine performs recipient resolution, deduplication, priority assignment, and persistence before UI delivery.

18.8 User Login

flowchart TD
A[User Submits Credentials] --> B[Validate Input]
B --> C[Find User by Email]
C --> D{User Exists and Active?}
D -- No --> E[Reject Authentication]
D -- Yes --> F[Verify Password Hash]
F --> G{Valid?}
G -- No --> H[Increment Failed Counter]
H --> I[Return 401]
G -- Yes --> J[Issue JWT and Refresh Token]
J --> K[Record Login Audit]
K --> L[Return Session Payload]

Login is stateless from the API perspective, with JWT-based access and refresh token rotation. Disabled users are not permitted to authenticate even with previously valid credentials. Failed login attempts are auditable and rate-limited.

18.9 eBay Account Onboarding

flowchart TD
A[Manager Clicks Connect eBay Account] --> B[Redirect to eBay OAuth]
B --> C[User Grants Consent]
C --> D[eBay Returns Authorization Code]
D --> E[Backend Exchanges for Tokens]
E --> F[Encrypt and Store Tokens]
F --> G[Test Account Access]
G --> H{Successful?}
H -- No --> I[Mark Account Error]
H -- Yes --> J[Set Status CONNECTED]
J --> K[Schedule Initial Sync]
K --> L[Write Audit Event]

Onboarding requires manager or admin authority. The platform stores encrypted refresh credentials, validates the new account connection, and schedules an initial sync. If validation fails, the account remains in an error state pending corrective action.

19. Security Requirements

ID	Control Area	Requirement
SEC-001	Authentication	Access tokens shall be JWTs signed by the backend with short TTL; refresh tokens shall be stored server-side.
SEC-002	Password Storage	User passwords shall be hashed using bcrypt with appropriate work factor and never logged in plaintext.
SEC-003	Authorization	RBAC shall be enforced on every protected endpoint and on sensitive UI controls.
SEC-004	Transport	All traffic shall use HTTPS/TLS 1.2+ in non-local environments.
SEC-005	Token Security	eBay refresh tokens shall be encrypted at rest using application-managed encryption keys.
SEC-006	Rate Limiting	Login, password reset, and reply-send endpoints shall be rate-limited to reduce abuse and brute force attempts.
SEC-007	Validation	All incoming payloads shall be schema-validated using Pydantic/FastAPI request models.
SEC-008	PII Handling	Logs and analytics exports shall mask or minimize buyer PII where full detail is not operationally required.
SEC-009	Secrets	Secrets shall be injected via environment variables or secret stores; no credentials stored in source control.
SEC-010	Dependency Hygiene	Container images and Python/Node dependencies shall be scanned regularly for critical vulnerabilities.

Because the system handles employee credentials, seller marketplace tokens, and buyer communication data, least privilege and security logging are mandatory. The application must also implement secure headers, CORS allowlists, refresh-token revocation on logout/password reset, and validation against malicious content. The MVP is token-based, so CSRF exposure is limited when not using cookie authentication.

20. Audit Requirements

Auditability is a first-class platform requirement because the system changes customer-facing communication state, operational ownership, and account connectivity. Audit events must be append-only, queryable, and role-restricted.

Audit Requirement	Description
Mandatory Payload	event_type, entity_type, entity_id, actor_user_id, occurred_at_utc, old_values, new_values, correlation_id
Minimum Event Set	LOGIN, LOGOUT, PASSWORD_RESET, USER_CREATED, USER_DISABLED, ACCOUNT_CONNECTED, ASSIGNMENT_CHANGED, CATEGORY_CHANGED, REPLY_SENT, REPLY_FAILED, TEMPLATE_USED
Immutability	Audit rows shall not be updated except by controlled redaction process approved by Admin.
Retention	Minimum 12 months recommended for MVP; production policy to be finalized by compliance owner.
Searchability	Admin and Operations Manager can filter by event type, user, date range, entity type, entity ID.

21. Reporting & Analytics

The reporting module provides operational visibility into message volume, queue health, agent productivity, and response behavior. Phase 1 delivers tactical dashboards, while Phase 2 expands to analytical trend views, category analysis, and SLA tracking.

KPI	Formula / Definition	Visible To
Open Conversations	Count of conversations where status = OPEN or PENDING	Manager, Admin
Assigned Conversations	Count of conversations with non-null current assignee	All (role-filtered)
Resolved Conversations	Count of conversations where status = RESOLVED within selected period	Manager, Admin
Average First Response Time	Average of first_agent_reply_at - first_buyer_message_at	Manager, Admin
Resolution Rate	Resolved conversations / total active conversations within range	Manager, Admin
Agent Workload	Open assigned conversations by agent at time of query	Manager, Admin

Filters must include date range, eBay account, category, assignee, and status. CSV export is required in Phase 2; PDF export is optional for MVP and may be delivered later if the reporting library footprint is acceptable.

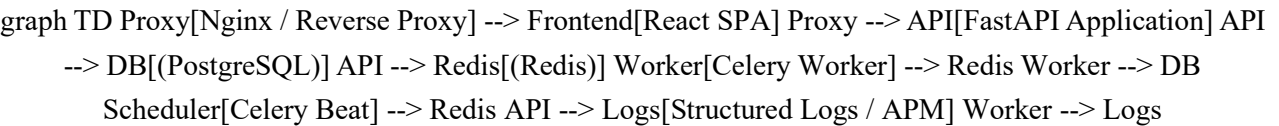
22. Notifications

Type	Trigger	Priority	Target Users
New Assignment	Conversation assigned by another user	Medium	New Assignee
Failed Reply	Outbound reply exhausts retries and remains FAILED	High	Assignee, Manager
Account Health Warning	Repeated sync failure or token expiry approaching threshold	High	Manager, Admin
Future Mention	@username in internal note (Phase 3)	Medium	Mentioned User

Notifications are in-app for MVP. Each record supports unread/read state, created timestamp, actor reference, target reference, deduplication key, and optional deep link to the affected conversation or account. Delivery must be durable even when the user is offline. Future enhancement: email and collaboration channel integrations.

23. Deployment Architecture

The MVP deployment model uses Docker Compose to support local development, UAT, and a lightweight production-like environment. While Compose is not a long-term orchestration target for high-scale production, it is sufficient for the expected volume and team size in the initial rollout.



Environment	Purpose	Notes
Local	Developer setup	Compose services: frontend, api, db, redis, worker, beat; seeded data optional
UAT	Business validation and QA signoff	Near-production config, sanitized test accounts, persistent volumes
Production-like	Operational deployment baseline	Backups enabled, TLS terminated at proxy, health checks enforced

Minimum health checks: API `/health`, database readiness, Redis readiness, worker heartbeat, and last sync age threshold. PostgreSQL backups should run on a scheduled basis with restore rehearsals. Alembic migrations must be applied as a controlled startup step in deployment pipelines.

24. Acceptance Criteria

24.1 UAT Criteria

ID	Scenario	Acceptance Criterion
UAT-001	User Login	Given valid credentials, when the user logs in, then the dashboard loads and a login audit event is created.
UAT-002	New Message Visibility	Given a new eBay buyer message, when the sync job completes, then a conversation is visible in the queue.
UAT-003	Reply Dispatch	Given an assigned conversation, when an agent sends a reply, then the message shows SENT or FAILED with retry history.
UAT-004	Assignment	Given an unassigned conversation, when a manager assigns it to an agent, then the assignee receives a notification.
UAT-005	Template Usage	Given a template exists, when an agent inserts and sends it, then template usage is reflected in audit logs.

24.2 Functional Test Cases

ID	Module	Test Case	Expected Result
FT-001	Authentication	Login with invalid password	401 Unauthorized returned; no session created; failed login recorded.
FT-002	User Management	Create user with duplicate email	409 Conflict / validation error; user not created.
FT-003	Conversation	Filter by assignee and status	Only matching conversations appear in queue.
FT-004	Reply	Send reply with duplicate idempotency key	Second request returns previous send result or conflict per API policy; no duplicate message sent.
FT-005	Notifications	Mark notification read	Unread count decreases and notification state persists on refresh.

24.3 Integration Test Cases

ID	Integration	Test Objective	Expected Result
IT-001	eBay OAuth	Complete account connection flow	Tokens exchanged, encrypted, stored, account marked CONNECTED.
IT-002	Sync + Persistence	Import message payload from eBay	Conversation/message rows created without duplicates.
IT-003	Reply + eBay API	Dispatch outbound reply and capture provider response	Status transitions from PENDING to SENT and provider reference stored.
IT-004	Audit + Business Event	Assignment change creates audit and history rows	AssignmentHistory and AuditLog rows both exist and correlate.

24.4 Performance Test Cases

ID	Scenario	Target
PT-001	Dashboard page load with representative data set	< 3 seconds
PT-002	Conversation list query at 25 concurrent users	P95 API latency < 500ms
PT-003	Scheduled sync for 400 messages/day workload	No backlog growth under nominal load
PT-004	Reply dispatch with transient provider errors	Retry logic behaves correctly; no duplicate sends

25. Future Enhancements

- **Phase 2:** AI categorization, first response SLA timers, resolution SLA tracking, breach alerts, agent productivity views, advanced trend analytics, and bulk queue operations.
- **Phase 3:** Internal notes, mention notifications, escalation rules, AI reply suggestions, custom reporting, and collaboration enhancements.
- **Marketplace Foundation:** Abstract marketplace adapters to support Amazon, Etsy, and Walmart by encapsulating provider-specific authentication, sync, and reply APIs behind a common interface.
- **Operational Maturity:** Enhanced observability, long-term archival strategies, and optional event-driven streaming architecture if throughput materially increases.

26. Traceability Matrix

Business Goal	Module / Epic	Requirement IDs	API / Interfaces	Entities	Test Coverage
Centralize Support Operations	Message Sync + Conversations	FR-SYNC-001, FR-CONV-001, FR-CONV-002	/api/v1/conversations, scheduled sync workers	Conversation, Message, EbayAccount	UAT-002, IT-002, PT-003
Reduce Response Times	Reply Management + Templates	FR-REPLY-001, FR-REPLY-002, FR-TPL-001	/api/v1/conversations/{id}/replies, /api/v1/templates	Message, Template, AuditLog	UAT-003, FT-004, IT-003
Eliminate Account Switching	eBay Account Management	FR-EBY-001, FR-EBY-002, FR-EBY-003	/api/v1/ebay-accounts*	EbayAccount, MessageSyncJob	IT-001
Improve Team Visibility	Dashboard + Notifications	FR-DASH-001, FR-NOTIF-001	/api/v1/analytics/overview, /api/v1/notifications	Notification, Conversation	UAT-004, FT-005, PT-001
Track Ownership of Conversations	Assignment & History	FR-ASG-001, FR-ASG-002, BR-001	/api/v1/conversations/{id}/assign	Conversation, AssignmentHistory, AuditLog	UAT-004, IT-004
Provide Management Reporting	Analytics	FR-DASH-002, FR-SLA-001	/api/v1/analytics/overview, /api/v1/analytics/agents	Conversation, Message, User	PT-001, PT-002
Maintain Complete Communication History	Audit + Message Persistence	FR-AUD-001, FR-SYNC-002, BR-004	/api/v1/audit-logs	AuditLog, Message, Conversation	UAT-005, IT-004